

Chapter	Title	Total periods	Duration
Ch 8	Journey Inside the Atom	11	50 mins × 7 · 60 mins × 4

C No.	Targeted NCF competencies	Justification
C-1.1	Describes classification of elements in the Periodic Table, and explains how compounds (including carbon compounds) are formed based on atomic structure (Bohr's model) and properties (valency)	The chapter's whole domain is atomic structure via the Bohr model and the combining capacity (valency) that follows from electronic configuration — the 'atomic structure (Bohr's model) and properties (valency)' clause of C-1.1. The periodic-table / compound-formation clause of C-1.1 is not developed here; it is the domain of the sibling chapter (Atomic Foundation of Matter). The chapter's four sub-topics (model evolution, atomic/mass number and electronic configuration, valency, isotopes/isobars) are all facets of atomic structure and serve this single competency; no distinct second strand exists within CG-1 (mixtures C-1.2 and chemical equations C-1.3 are absent).

This assessment is formative. Every question traces to a specific learning outcome, and each outcome is derived from an NCF competency this chapter develops. Rather than testing recall or asking whether a student “understood” a topic, each item asks for an observable demonstration — a response, a classification, a justification, or a product — of the thinking the lessons built. The questions are grouped by sections, and each shows the learning outcome it assesses and its cognitive demand, so you can see exactly what each question checks and why.

SECTION 1

8.1 Rediscovering the Roots of Atomic Theory

Period 1	50 min	Tracing the First Atomic Ideas from Kanada to Dalton
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Materials: Blackboard or whiteboard, Chalk or markers, Textbook Chapter 8

Teacher notes: This is the chapter's entry point, so orient students to the broad arc: they will trace how ideas about atoms evolved from philosophy to experimental science over two thousand years. Watch for the misconception that ancient Indian and Greek thinkers were doing science the same way Dalton was — the key distinction is the absence of experimental evidence in the earlier accounts. A useful checkpoint: ask students to state in one sentence what Dalton added that the ancient thinkers lacked; if they can articulate 'experimental evidence and specific claims about combining ratios,' they are ready to move on.

8 min	Open by asking students where they think the idea of smallest particles of matter came from. Take two or three responses, then establish that independent thinkers in ancient India and ancient Greece arrived at nearly identical ideas without contact with each other.
14 min	Present Acharya Kanada's proposal from the Vaisesika Sutras — matter (dravya) divided repeatedly yields parmanus, the indivisible units that combine in dyads and triads to form all substances. Draw the Sanskrit term on the board alongside the Greek atomos from Leucippus and Democritus. Emphasise: both traditions arrived at this through philosophical reasoning, not experiment.
16 min	Move to Dalton's 1808 scientific treatment. Ask students: what changed between the Greek idea and Dalton's proposal? Guide them to identify the shift from philosophical argument to scientific claims about elements and compounds. Students write in their notebooks: two similarities and one key difference between the ancient philosophical atoms and Dalton's atoms.
12 min	Pairs share their comparisons. Consolidate on the board: both traditions posit indivisibility; Dalton added the claim that different elements have different atoms and that atoms combine in fixed ratios. Close by posing the question the next sections will answer: if atoms are indivisible, how do we explain the existence of electrons?

Homework: Write two sentences explaining why Dalton's atomic theory marked a shift from the earlier philosophical ideas about atoms, using a specific difference you identified in class.

Assessment

Q1 · Learning Outcome

Students can distinguish between philosophical reasoning and evidence-based scientific argument by identifying the specific evidential criteria that one framework provides and the other lacks.

Cognitive Demand
Understanding

Acharya Kanada and the Greek philosopher Democritus both proposed that matter consists of tiny, indivisible particles. Dalton, in 1808, also proposed indivisible atoms. What does Dalton's proposal have that the ancient accounts lack?

- A. Dalton gave atoms specific names derived from Sanskrit and Greek roots.
- B. Dalton was the first to propose that matter is made of indivisible particles, while Kanada and Democritus did not.
- C. Dalton's atoms are based on experimental evidence and specific claims about how atoms of different elements combine in fixed ratios. ✓
- D. Dalton's atoms are spherical in shape, whereas ancient philosophers described atoms as having different geometrical forms.

Answer section

Correct answer: C

What each choice reveals

- A. Conflates naming conventions with the nature of the evidential basis — the distinction the question tests is philosophical vs experimental, not etymological origin.
- B. Directly contradicts the chapter content, which states both ancient traditions proposed indivisible particles independently before Dalton — selecting this reveals a factual misread of the section.
- D. Introduces a shape detail not present in section 8.1 and misidentifies what changed between ancient and Daltonian atomism — the change is in evidential grounding, not geometry.

SECTION 2

8.2 A Short Historical Journey Through Atomic Models

Period 2 60 min Discovering the Electron and Building Thomson's Plum Pudding Model

Materials: Textbook Chapter 8, Blackboard or whiteboard, Chalk or markers

Teacher notes: Having established that Dalton's indivisible atom was the scientific starting point, this unit shows students how the discovery of the electron broke that assumption. A common error is thinking that Thomson's cathode ray experiment proved the atom has internal structure on its own — students should see that the neutrality constraint is what forced Thomson to propose the plum pudding arrangement. Encourage students to engage seriously with the analogy evaluation task rather than treating it as decorative; asking 'what physical property of a lemon differs from a watermelon that matters for this model?' can sharpen their thinking.

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| 10 min | Pose the problem: Dalton said atoms are indivisible. Then how do we know they contain smaller parts? Ask students to speculate what kind of evidence might force a change in that idea. Take responses and note them on the board without evaluating yet. |
| 17 min | Present Thomson's 1897 cathode ray tube experiment. Draw the tube on the board: cathode, anode, low-pressure gas, deflection plates. Explain that the rays bent toward the positive plate (so they carry negative charge), and were the same regardless of the cathode material (so they must be universal). Students note in their books: two pieces of evidence and the inference Thomson drew from each. |
| 16 min | Introduce the problem Thomson faced: electrons are negative, but atoms are neutral. Where is the positive charge? Present the plum pudding model — a uniform positive sphere with electrons distributed throughout, like seeds in a watermelon. Students sketch and label the model in their notebooks, then answer in writing: what does the watermelon analogy get right, and what does it leave unexplained? |

17 min	Facilitate a brief discussion on the Pause and Ponder question from the chapter: would a lemon or an orange be a better analogy than a watermelon, and where does each comparison break down? Students articulate one strength and one limitation of each analogy. Consolidate: good models fit the data we have but leave open questions that new experiments must answer.
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Assessment

Q2 • Learning Outcome

Students can evaluate a scientific model by identifying the evidence that supports it and the questions it leaves unresolved, using an analogy to expose the model's structural assumptions.

Cognitive Demand Analysis

Thomson's cathode ray experiment provided two key pieces of evidence that led him to propose the plum pudding model of the atom.

- (a) State the two pieces of evidence and the inference each one supports.
- (b) The plum pudding model is often compared to a watermelon — red pulp as positive matter, seeds as electrons. Identify one aspect of atomic structure the watermelon analogy represents correctly and one structural question about the atom the analogy cannot answer.

Answer section

Look for

- Correctly identifies both pieces of evidence: cathode rays deflect toward the positive plate (negative charge) AND cathode rays are the same regardless of cathode material (electrons are universal).
- Links each piece of evidence to its specific inference rather than restating it: deflection → electrons are negatively charged; independence from material → electrons are present in all elements.
- Correctly identifies the watermelon analogy's strength: it represents the coexistence of positive and negative components in a single neutral structure.
- Correctly identifies the analogy's limitation: it cannot explain where the positive charge comes from, what holds the electrons in specific positions, or how the model would be tested experimentally — any one of these counts.
- Reasoning is explicit: the student states why each inference follows from the evidence, not just what the inference is.

SECTION 3

8.2.2 Testing Thomson's model: The gold foil experiment

Period 3

50 min

The Gold Foil Experiment and What Backscattering Revealed

Materials: Textbook Chapter 8 (gold foil diagram), Blank paper for student prediction diagrams, Pencils

Teacher notes: Having built Thomson's plum pudding model in the previous unit, this unit uses the gold foil experiment to demonstrate the scientific process of model testing and revision. The most common misconception is that 'most particles passed through' supports Thomson's model — students must understand that it is the few that bounced back that shattered it. Asking 'would Thomson's model predict even a single particle bouncing back?' focuses students on the critical observation.

10 min	Display the gold foil experimental diagram. Ask students: if Thomson's plum pudding model were correct — positive charge spread evenly through the atom — what deflection pattern would you predict for alpha particles fired at the foil? Students sketch their prediction diagram individually before any discussion.
15 min	Reveal the actual results: most particles pass straight through, some deflect at large angles, and a few bounce directly back. Students compare their prediction to the actual outcome and write one sentence naming the specific observation that their prediction failed to explain. Introduce the term 'scattering' and the alternate name 'alpha-ray scattering experiment.'
13 min	Guide students through the Think as a Scientist task: if the gold foil were made thicker, how would the scattering pattern change? Students draw a modified diagram and explain in writing why increased thickness raises the probability of an alpha particle encountering a nucleus. Share and compare diagrams in pairs.
12 min	Address the Pause and Ponder question: a single observation — sharp backscattering of a few particles — is enough to rule out Thomson's model entirely. Students write a two-sentence logical argument explaining why this one result is fatal to the plum pudding picture, then share with a neighbour. Consolidate: in science, one contradicting result is sufficient to overturn a model.

Homework: Draw the gold foil experimental setup from memory and label: alpha particle source, gold foil, and the three categories of particle paths. Add one sentence explaining what each category of path tells us about atomic structure.

Period 4

50 min

Rutherford's Planetary Model, Its Instability, and the Proton

Materials: Blackboard or whiteboard, Chalk or markers, Textbook Chapter 8

Teacher notes: Having established the gold foil results in the previous unit, this unit draws out Rutherford's three conclusions and their logical limits. A frequent confusion is between the reason the atom is mostly empty (explains undeflected particles) and the reason there must be a dense nucleus (explains backscattering) — keep these two inferences distinct on the board. The assertion-reason task is a good formative checkpoint: students who conflate Thomson's and Rutherford's models will reveal it here.

10 min	Review the three key conclusions Rutherford drew from the gold foil data: most of the atom is empty space; all positive charge and most mass concentrate in a tiny nucleus; electrons orbit the nucleus. Build the planetary model diagram on the board and establish the size comparison — nucleus is about 10^5 times smaller than the atom, comparable to a pepper grain at the centre of a cricket ground.
15 min	Present the stability problem. Ask: according to classical physics, a charged particle moving in a circle is constantly accelerating. What must it therefore be doing? Guide students to the inference: it must radiate energy, slow down, and spiral into the nucleus. Students write in their notebooks: why Rutherford's model predicts atomic collapse and why this is a problem.
13 min	Introduce the discovery of the proton: Rutherford showed the nucleus carries positive charge from protons — heavier than electrons, charge +1. For electrical neutrality, number of protons must equal number of electrons. Students apply this to helium (2p, 2e) and sodium (11p, 11e), writing the subatomic composition for each.
12 min	Students work the Pause and Ponder assertion-reason question: evaluate whether Rutherford's conclusion about nuclear mass concentration is correctly explained by Thomson's model of uniformly distributed positive charge. Students write a structured argument (claim, evidence, reasoning) and share in pairs. Consolidate: Rutherford's model explains the scattering but fails to explain stability — the unresolved question Bohr will answer.

Assessment

Q3 · Learning Outcome

Students can construct a logical argument showing how a single anomalous experimental result falsifies a model, and derive structural inferences about a system from the pattern of observed deflections.

Cognitive Demand Analysis

In the gold foil experiment, most alpha particles passed straight through the foil, a small number deflected at large angles, and a very few bounced directly back. Which observation is sufficient on its own to falsify Thomson's plum pudding model, and what structural feature of the atom does it imply?

- A. A few particles bouncing directly back — this implies a tiny, dense, positively charged nucleus, which Thomson's diffuse positive sphere cannot produce. ✓**
- B.** Most particles passing straight through — this shows the atom is mostly empty space, which Thomson's model did not predict.
- C.** Some particles deflecting at large angles — this shows electrons are negatively charged, confirming Thomson's placement of negative charges inside the atom.
- D.** The combination of all three observations together is required; no single observation alone is sufficient to falsify Thomson's model.

Answer section

Correct answer: A

What each choice reveals

- B** Confuses the inference from undeflected particles (empty space) with what falsifies Thomson's model — Thomson's model did predict slight deflections from diffuse charge, so undeflected particles alone are not sufficient to falsify it; the backscattering is what no diffuse model can explain.
- C** Misidentifies large-angle deflection as confirming Thomson's electron placement rather than challenging Thomson's positive-charge distribution; also incorrectly attributes large-angle deflection to electron behaviour rather than nuclear concentration.
- D** Misunderstands the logic of falsification — in science, one sufficiently anomalous observation falsifies a model; demanding all three together reflects a misunderstanding of how experimental evidence works against theoretical predictions.

Q4 · Learning Outcome

Students can construct a logical argument showing how a single anomalous experimental result falsifies a model, and derive structural inferences about a system from the pattern of observed deflections.

Cognitive Demand

Application

Rutherford's planetary model drew three conclusions from the gold foil experiment. However, a physics principle requires a charged particle moving in a circular orbit to continuously radiate energy.

- (a) State the three conclusions Rutherford drew.
- (b) Use the physics principle stated above to explain why an electron orbiting the nucleus according to Rutherford's model would be unstable.
- (c) A helium atom has 2 protons. State the number of electrons in a neutral helium atom and explain why.

Answer section**Expected elements**

- Three conclusions: most of the atom is empty space; all positive charge and most mass are concentrated in a tiny nucleus; electrons revolve around the nucleus in orbits.
- Instability argument: because the orbiting electron is accelerating (circular motion), it must radiate energy; losing energy causes it to spiral inward and collapse into the nucleus.
- Helium has 2 electrons in the neutral atom; the number of electrons equals the number of protons because the atom is electrically neutral (charges must balance).

SECTION 4**8.2.3 Bohr's model of the atom****Period 5**

60 min

Bohr's Shell Model and Why Electrons Do Not Spiral Inward

Materials: Blackboard or whiteboard, Chalk or markers, Textbook Chapter 8

Teacher notes: Having established Rutherford's stability problem, this unit shows how Bohr's stationary-state postulate resolves it. A persistent misconception is that electrons in Bohr's model are stationary — clarify that they are in motion but do not radiate because they are in a stationary state defined by a fixed energy. If students ask why electrons do not radiate in a stationary state, acknowledge this as a postulate Bohr introduced without classical justification, and that the full explanation requires quantum mechanics studied later.

12 min	Recall Rutherford's collapse problem — orbiting electrons must radiate energy and spiral in. Present Bohr's 1913 solution: electrons occupy fixed circular paths called stationary states or shells, each with a definite constant energy. In a stationary state, the electron does not lose energy despite its motion. Students note this as the key resolution of Rutherford's problem.
18 min	Present the shell labelling system: K ($n=1$, lowest energy, closest), L ($n=2$), M ($n=3$), N ($n=4$). Draw concentric circles and label them on the board. Share the Threads of Curiosity explanation for why letters K-N are used rather than A-D: Charles Barkla's X-ray convention reserved lower letters in case earlier series were found. Students note both the labels and the rationale.
16 min	Explain electron transitions: an electron moves from a lower shell to a higher shell only by absorbing energy exactly equal to the difference between the two levels; it returns by emitting that same quantity of energy. Draw a transition arrow on the board. Students write two sentences contrasting what happens to an electron in Rutherford's model versus in Bohr's model during motion.
14 min	Students examine a board diagram of the Bohr model and identify: which shell has the lowest energy, in which direction energy increases, and what condition allows an electron to jump to a higher shell. Students answer three short written questions individually, then compare with a partner. Consolidate: Bohr's model resolves stability but still leaves the question of how many electrons each shell can hold.

Homework: Draw Bohr's shell model for an atom with electrons in the K and L shells. Add labels showing which shell has lower energy and draw an arrow showing an electron absorbing energy to move outward. Write one sentence explaining what happens to the electron if it later releases that energy.

Assessment

Q5 · Learning Outcome

Students can explain how introducing the concept of fixed energy levels resolves a contradiction in a prior model by identifying the specific physical consequence the new postulate prevents.

Cognitive Demand

Understanding

Bohr proposed that electrons occupy fixed circular paths called stationary states, each with a definite constant energy. Which statement correctly explains how this postulate resolves the problem with Rutherford's model?

- A. Bohr's shells are labelled K, L, M, N because earlier letters were reserved for a possible inner series, which prevents energy loss from the atom.
- B. Electrons in Bohr's model can only move between shells when they absorb energy, so they naturally remain in the lowest shell and never spiral inward.
- C. **In a stationary state, the electron moves but does not radiate energy, so it maintains its orbit indefinitely without spiralling into the nucleus. ✓**
- D. In a stationary state, the electron stops moving, so it no longer accelerates and does not radiate energy.

Answer section

Correct answer: C

What each choice reveals

- D Conflates 'stationary state' with the electron being literally stationary — a common and important misconception; students selecting this have not distinguished between the electron's motion and its energy state.
- A Confuses the historical origin of the K-N labelling convention (an interesting but structurally unrelated fact) with the physical mechanism that prevents energy loss — a confusion between a naming convention and a physical postulate.
- B Partially correct (electrons do need to absorb energy to move to a higher shell) but explains why electrons stay in one shell by appeal to the absorption rule rather than the stationary-state postulate; it also implies an electron cannot be in any shell above K by default, which is wrong for multi-electron atoms.

SECTION 5**8.3 What Components Contribute to the Mass of an Atom?****Period 6**

50 min

The Neutron Discovery and Solving the Nuclear Mass Puzzle

Materials: Blackboard or whiteboard, Chalk or markers, Textbook Chapter 8

Teacher notes: Having established that Bohr's model explained electron stability, this unit shifts attention to nuclear composition and resolves the mass discrepancy that protons alone could not explain. A frequent error is thinking neutrons are just extra protons without charge — emphasise that neutrons are genuinely distinct particles with a separate role in binding the nucleus. The hypothesis-writing task at the opening is important: it surfaces students' prior ideas and creates a genuine moment of revision when Chadwick's solution is introduced.

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| 10 min | Pose the mass puzzle on the board: hydrogen has 1 proton; helium has 2 protons. If only protons contribute to nuclear mass, helium should be twice as heavy as hydrogen. But helium is actually about four times heavier. Ask students to write a hypothesis in one sentence explaining where the extra mass might come from, without telling them the answer. |
| 15 min | Present Chadwick's 1932 discovery of the neutron: a particle with mass nearly equal to a proton but carrying no electrical charge (symbol n^0). Chadwick was a student of Rutherford. Neutrons are found in the nucleus of all atoms except ordinary hydrogen. Draw the neutron into the board nucleus diagram. Students now resolve their earlier hypothesis: was their prediction correct or did it need revision? |
| 13 min | Build the three-particle comparison table on the board: electron (e^- , charge -1 , negligible mass), proton (p^+ , charge $+1$, mass $\approx 1u$), neutron (n^0 , charge 0 , mass $\approx 1u$). Students copy and complete the table. Then address nuclear stability: ask why do you think heavier nuclei need more neutrons? Guide students toward the idea that protons repel each other, so neutrons are needed to provide the binding force and dilute repulsion. |
| 12 min | Present the uranium example: 92 protons, 146 neutrons. Students work the What If question from the chapter: how would the size of everyday objects change if atoms had no empty space? Students write two sentences reasoning about this, then share. Consolidate: the neutron resolves the mass discrepancy and plays a structural role in nuclear stability that protons alone cannot provide. |

Assessment

Q6 · Learning Outcome

Students can resolve a quantitative discrepancy in a scientific model by identifying the missing component and explaining the physical role that component plays in the system's stability.

Cognitive Demand

Application

Hydrogen has 1 proton and helium has 2 protons, yet helium is approximately four times as heavy as hydrogen, not twice as heavy.

- (a) Identify the particle discovered by Chadwick that accounts for this discrepancy and state its charge and approximate mass.
- (b) Uranium has 92 protons and 146 neutrons. Explain, using the information about neutrons from this chapter, why uranium needs far more neutrons than protons to remain stable.

Answer section**Expected elements**

- Neutron: charge 0 (neutral), mass approximately equal to a proton ($\sim 1u$); found in the nucleus.
- Helium has 2 protons and 2 neutrons, giving mass $\approx 4u$, resolving the discrepancy — mass comes from both protons and neutrons.
- Protons repel each other electrostatically; neutrons have no charge so they dilute the repulsion by spacing protons apart and strengthen the nuclear binding force — heavier nuclei need progressively more neutrons to counterbalance the increased proton repulsion.

SECTION 6**8.4 Symbols of Elements****Period 7**

50 min

Element Symbols, Atomic Number, and Mass Number in Notation

Materials: Textbook Chapter 8 (27-element reference table), Blank paper or notebook, Pencils

Teacher notes: This unit covers two sections — symbols (8.4) and atomic number and mass number (8.5 and 8.6) — because their combined content forms a single coherent notation system most naturally practised together. A very common error is confusing which number goes in the superscript and which in the subscript; reinforce the mnemonic that the subscript Z is 'below' because protons are inside the nucleus, and mass (A) sits 'above' as the total count. The hands-on symbol-invention task at the opening is intentionally placed before the rules are taught — students who invent an all-caps symbol or a symbol with a number will self-correct when they apply the IUPAC criteria.

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| 10 min | Open with the Pause and Ponder task from section 8.4: students invent a name and symbol for a newly discovered element. They work individually, then apply the IUPAC rules from the chapter to evaluate whether their own symbol is valid. Pairs compare and identify which symbols violate the rules and why. |
| 12 min | Consolidate the IUPAC symbol rules on the board: first letter capitalised, second letter (if used) lowercase, may use non-adjacent letter, some symbols from Latin/Greek/German names. Work through examples: Fe (ferrum), Hg (hydrargyros), W (wolfram), Cl (chlorine). Students match five symbols from the reference table to their derivation and write the rule each illustrates. |
| 13 min | Introduce atomic number (Z) — number of protons in the nucleus, equal to the number of electrons in a neutral atom, the unique identity of each element. Students write the atomic number and electron count for hydrogen ($Z=1$), helium ($Z=2$), and lithium ($Z=3$). Then introduce mass number (A) — total nucleons (protons + neutrons). Present the standard notation with A as superscript and Z as subscript. Work through ${}^1\text{H}$, ${}^4_2\text{He}$, ${}^7_3\text{Li}$ on the board. |
| 15 min | Students practise a quick individual calculation set: for each of four given nuclide notations, determine the number of protons, electrons, and neutrons. They then construct the correct notation for an element given its proton count and mass number. Students check each other's work in pairs. Consolidate: the relationship $A - Z =$ neutron count is the core calculation tool. |

Homework: For each of the following elements — carbon ($Z=6$, $A=12$), fluorine ($Z=9$, $A=19$), and sodium ($Z=11$, $A=23$) — write the standard nuclide notation and state the number of neutrons.

Assessment**Q7 · Learning Outcome**

Students can apply a rule-governed symbolic notation system to encode and decode quantitative information about atomic composition from a standardised representation.

Cognitive Demand

Application

The nuclide notation for an atom is written as $^{23}_{11}\text{Na}$. Using this notation, how many neutrons does this atom contain?

- A. 11
- B. 12 ✓
- C. 23
- D. 34

Answer section

Correct answer: B

What each choice reveals

- A Reads the subscript ($Z=11$) as the neutron count, confusing atomic number with neutron number — the student has not applied $A-Z$.
- C Reads the superscript ($A=23$) as the neutron count, confusing mass number with neutron number — the student has not subtracted Z from A .
- D Adds A and Z ($23+11=34$) instead of subtracting — a procedural error revealing the student knows both numbers are relevant but has applied the wrong operation.

SECTION 7
8.6 Mass Number

Period 8	60 min	Mass Number and Neutron Count — Numerical Practice
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Materials: Textbook Chapter 8, Notebook and pencils, Blackboard or whiteboard

Teacher notes: Having introduced standard nuclide notation in the previous unit, this unit gives students extended practice with the $A-Z$ relationship before moving to shell filling. The most common error is substituting the element's period number for Z , or confusing the mass number with atomic mass in atomic mass units — keep the two concepts clearly separated. A good formative check is to ask a student to explain why an element's number of neutrons can differ from its number of protons, which surfaces whether they have genuinely internalised the $A-Z=\text{neutrons}$ relationship.

12 min	Establish the mass number definition: A equals total nucleons (protons + neutrons); electron mass is negligible so A accounts for virtually all atomic mass. Work hydrogen ($A=1$, $Z=1$, neutrons=0), helium ($A=4$, $Z=2$, neutrons=2), lithium ($A=7$, $Z=3$, neutrons=4) on the board, each time deriving neutrons as $A-Z$.
24 min	Students work the Pause and Ponder numerical problems from section 8.6 individually: finding electrons, protons, and neutrons given atomic number and nucleon count; finding neutrons given atomic number and mass number; finding mass number given neutron count and atomic number. Students write each step explicitly (A , Z given $\rightarrow$ neutrons = $A-Z$).
14 min	Pairs exchange notebooks and check each other's working. Students identify and annotate any step where a partner's reasoning went wrong, not just the final answer. Reconvene and work one or two commonly missed problems on the board, making the reasoning chain visible.
10 min	Consolidate: A is the mass count, Z is the identity count, and $A-Z$ gives the neutrons. Students write in their own words a single rule that connects these three quantities, without using numbers, as a test of conceptual understanding rather than procedural recall.

Assessment

Q8 · Learning Outcome Students can apply a multi-variable numerical relationship to determine unknown subatomic quantities, showing each step of the reasoning chain explicitly.	Cognitive Demand Application
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Refer to the table provided. Complete all missing values for each atom, showing the relationship you use for each row.

Atom	Atomic Number (Z)	Mass Number (A)	Protons	Neutrons	Electrons
Carbon	6	12	?	?	?
Sulfur	16	32	?	?	?
Chlorine	17	?	17	18	?

Calcium	?	40	20	?	20
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Answer section

Expected elements

- Carbon: protons = 6 (equals Z), neutrons = $A - Z = 12 - 6 = 6$, electrons = 6 (equals Z for neutral atom).
- Sulfur: protons = 16, neutrons = $32 - 16 = 16$, electrons = 16.
- Chlorine: $A = \text{protons} + \text{neutrons} = 17 + 18 = 35$; electrons = 17 (equals Z).
- Calcium: $Z = 20$ (equals proton count given); neutrons = $A - Z = 40 - 20 = 20$.
- Student explicitly states the relationship used for each calculation ($A - Z$ for neutrons; Z equals electrons for neutral atom; $A = \text{protons} + \text{neutrons}$ for mass number).

SECTION 8

8.7 How Are Electrons Distributed in Different Energy Levels?

Period 9

50 min

Shell-Filling Rules and Electronic Configurations for the First Eighteen Elements

Materials: Textbook Chapter 8 (Fig. 8.11 and configuration table), Blank paper for drawing atomic structures, Pencils or pens, Blackboard

Teacher notes: Having established that Bohr's shells have fixed energies, this unit applies the $2n^2$ rule to actually populate atoms. The most persistent error is placing more than 8 electrons in the outermost shell even when the $2n^2$ formula would permit more — the outermost-shell cap of 8 (exception: K at 2) must be stated explicitly and checked at every step. Asking students to check neon ($Z=10$) and argon ($Z=18$) against both rules simultaneously is a good diagnostic: if a student writes 10 in the L shell for neon they have missed the outermost-shell constraint.

10 min	Present the three Bohr-Bury rules on the board: maximum electrons in shell n is $2n^2$ ($K \rightarrow 2$, $L \rightarrow 8$, $M \rightarrow 18$); outermost shell holds no more than 8; electrons fill shells in order from the innermost. Students copy the rules and calculate the capacity of K, L, and M shells using the formula, confirming the values.
18 min	Build electronic configurations step by step from hydrogen ($Z=1$) to argon ($Z=18$) on the board, adding one electron per element. At each step ask students: which shell does the next electron enter? Students draw 2-D schematic atomic structures for at least six elements of their choice from this range in their notebooks, labelling each shell with its electron count.
12 min	Students work the Pause and Ponder tasks from section 8.7: identify the number of valence electrons for $^{12}_6\text{C}$, $^{19}_9\text{F}$, and $^{28}_{14}\text{Si}$ from their nuclide notations; write complete electronic configurations for elements with atomic numbers 12, 16, and 18. Students check their configurations against the textbook table.
10 min	Students attempt the sodium riddle from the Pause and Ponder: identify an element from its mass number and described properties. Students write the identification and the step-by-step reasoning. Consolidate by confirming neon (2,8) and argon (2,8,8) as examples of filled outermost shells.

Homework: Write the electronic configuration for phosphorus ($Z=15$) and chlorine ($Z=17$), showing the electron count in each shell. State the number of valence electrons for each element.

Assessment

Q9 · Learning Outcome

Students can construct the electron shell distribution of an atom by systematically applying capacity and filling-order rules, and identify the valence shell from the resulting configuration.

Cognitive Demand

Application

An atom has atomic number 16.

- (a) Write the electronic configuration of this atom, showing the number of electrons in each shell.
- (b) Identify the valence shell and state the number of valence electrons.
- (c) A student writes the configuration of the atom with atomic number 18 as 2, 8, 8. Another student writes it as 2, 16. State which is correct and explain the rule that makes the second configuration impossible.

Answer section

Expected elements

- Atomic number 16: K=2, L=8, M=6. Configuration 2, 8, 6.
- Valence shell is M; 6 valence electrons.
- Configuration 2, 8, 8 is correct for Z=18; the second (2, 16) is impossible because the outermost shell cannot hold more than 8 electrons regardless of its $2n^2$ capacity.

SECTION 9

8.8 Combining Capacity of an Atom: Valency

Period 10

60 min

Valence Electrons, the Octet Rule, and Predicting Valency

Materials: Textbook Chapter 8 (Table 8.4), Blackboard or whiteboard, Notebooks and pencils

Teacher notes: Having built electronic configurations for the first eighteen elements, this unit connects those configurations to chemical behaviour through the octet rule. A common error is treating valency as simply the number of valence electrons — students must distinguish between elements that lose electrons to empty the valence shell (like sodium with 1) versus those that gain electrons to fill it (like chlorine with 7, valency 1). Asking 'why does sodium lose rather than gain 7 electrons?' probes this distinction and usually surfaces the confusion.

12 min	Pose the inquiry question: atoms are electrically neutral, yet they react and combine. What drives an atom to interact with another? Ask students to hypothesise based on what they know about electronic configurations. Take two or three responses, note them on the board, and return to evaluate them at the end of the unit.
22 min	Introduce the octet rule: atoms with 8 valence electrons (or 2 for helium) are stable and largely unreactive. Atoms with incomplete valence shells achieve stability by losing, gaining, or sharing electrons. Work through sodium (2,8,1 → loses 1, valency 1), oxygen (2,6 → gains 2, valency 2), and carbon (2,4 → shares 4, valency 4) on the board. Students write the electronic configuration and valency determination for each.
14 min	Students apply the method independently: determine the valency of magnesium (Z=12), chlorine (Z=17), and nitrogen (Z=7) by first writing the electronic configuration, then identifying how many electrons are needed to complete or empty the valence shell. They record: configuration → valence electron count → action (lose/gain/share) → valency.
12 min	Return to the opening hypotheses on the board. Students evaluate each one against the octet rule: which hypotheses were correct, which were incomplete, which were wrong? Students extend Table 8.4 from the textbook by adding a valency column for the elements they have determined. Consolidate: valency 0 for noble gases; the chapter notes that some compounds violate simple valency rules, to be explored later.

Homework: Determine the valency of aluminium (Z=13), sulfur (Z=16), and argon (Z=18) by writing the electronic configuration for each and stating whether electrons are lost, gained, or shared. Explain in one sentence why argon has valency 0.

Assessment

Q10 • Learning Outcome

Students can predict the combining capacity of an atom by applying the octet rule to its electronic configuration and distinguishing between electron loss, gain, and sharing pathways.

Cognitive Demand

Application

Magnesium has atomic number 12. Its electronic configuration is 2, 8, 2. What is the valency of magnesium and what does the atom do to achieve a stable configuration?

- A. Valency 2; magnesium gains 2 electrons, filling its M shell to 8.
- B. Valency 2; magnesium loses 2 electrons, leaving a complete L shell as the new outermost shell. ✓**
- C. Valency 6; magnesium gains 6 electrons to fill its outermost shell to 8.
- D. Valency 8; magnesium shares 8 electrons to reach an octet.

Answer section

Correct answer: B

What each choice reveals

- C** Treats the 2 valence electrons as deficient by 6 (as if $8-2=6$ electrons must be added), confusing gaining with the number needed to fill — also incorrectly applies gaining rather than losing for an atom with only 2 valence electrons.
- A** Correctly identifies the valency as 2 but incorrectly assigns the action as gaining rather than losing; a student selecting this knows valency = 2 but has confused loss and gain for this configuration.
- D** Applies the number 8 (the target octet) as the valency rather than deriving valency as the number of electrons actually transferred or shared — a confusion between the target configuration and the action needed to reach it.

SECTION 10

8.9 A Deeper Look into Atomic Structure

Period 11

50 min

Isotopes, Isobars, and Weighted Average Atomic Mass

Materials: Textbook Chapter 8, Blackboard or whiteboard, Notebooks and pencils

Teacher notes: Having established electronic configuration and valency, this unit introduces isotopes and isobars as evidence that atoms of the same element are not always identical — extending beyond Dalton's original picture. The most persistent confusion is between isotopes and isobars: students mix up 'same Z different A' with 'same A different Z.' The comparison table task at the close is specifically designed to force this distinction into explicit form. The chlorine weighted average calculation deserves careful attention — students who interpret 35.5u as a property of a single atom have missed the statistical meaning of the weighted average.

- 10 min Open with the puzzle: if atomic number uniquely identifies an element, can two atoms of the same element have different masses? Ask students to reason about this before any instruction. Take hypotheses. Then introduce the definition: isotopes are atoms of the same element (same Z) with different mass numbers (different A) due to different neutron counts.
- 15 min Work through hydrogen's three isotopes on the board: protium (^1_1H , ~99.98%, 0 neutrons), deuterium (^2_1H , ~0.015%, 1 neutron), tritium (^3_1H , traces, 2 neutrons). Then carbon's three isotopes: $^{12}_6\text{C}$, $^{13}_6\text{C}$, $^{14}_6\text{C}$. Students note: same Z, same electron configuration, same chemical properties — different A, different physical properties. Students work the Pause and Ponder sodium isotope task: compare atomic numbers, mass numbers, and classify the pair as isotopes.
- 13 min Present the average atomic mass problem: chlorine has two isotopes $^{35}_{17}\text{Cl}$ (75%) and $^{37}_{17}\text{Cl}$ (25%). Simple average gives 36u, but the correct weighted average is $(35 \times 0.75) + (37 \times 0.25) = 35.5\text{u}$. Students work the bromine weighted average calculation from the Pause and Ponder task. Consolidate: 35.5u does not mean any atom has that mass — it reflects the isotopic mixture in nature.
- 12 min Introduce isobars: atoms of different elements with the same mass number but different atomic numbers. Example: calcium (Z=20), potassium (Z=19), argon (Z=18), all with A=40. Students construct a comparison table: isotopes vs. isobars — what is the same, what differs, and what that implies for chemical properties. The section closes by noting Bohr's model is itself incomplete — electrons form probability clouds rather than fixed orbits — with the quantum model treated in higher grades.

Homework: Calculate the weighted average atomic mass of boron, which has two isotopes: ^{10}B at 19.9% natural abundance and ^{11}B at 80.1% natural abundance. Show your working and state whether your answer represents the mass of any single boron atom, explaining why or why not.

Assessment

Q11 · Learning Outcome

Students can classify atomic variants by their conserved and differing quantities, and calculate a statistically weighted average to represent a naturally occurring mixture of isotopic masses.

Cognitive Demand
Application

Refer to the table provided.

- (a) Calculate the weighted average atomic mass of boron. Show your working.
- (b) Does any single boron atom in nature have the mass you calculated? Explain your answer.
- (c) Boron-10 and Boron-11 are isotopes of the same element. State two quantities that are the same for both atoms and one quantity that differs.

Isotope	Mass Number	Natural Abundance (%)
Boron-10	10	19.9

Boron-11	11	80.1
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Answer section

Expected elements

- Weighted average = $(10 \times 19.9/100) + (11 \times 80.1/100) = 1.99 + 8.811 = 10.8 \text{ u}$ (accept values rounded correctly to one decimal place).
- No single boron atom has mass 10.8 u; the value is a weighted average representing the isotopic mixture across a large population of boron atoms, not the mass of any individual atom.
- Same: atomic number ($Z=5$), number of electrons (5), chemical properties. Different: mass number (A), number of neutrons (5 vs 6). (Any two same and one different accepted.)

Q12 · Learning Outcome

Students can classify atomic variants by their conserved and differing quantities, and calculate a statistically weighted average to represent a naturally occurring mixture of isotopic masses.

Cognitive Demand Analysis

The chapter states that isotopes of an element have the same chemical properties but different physical properties. Argon ($Z=18$, $A=40$), potassium ($Z=19$, $A=40$), and calcium ($Z=20$, $A=40$) are described as isobars.

- (a) Explain, using the concept of electronic configuration, why the two isotopes of carbon (^{12}C and ^{13}C) have identical chemical properties despite having different masses.
- (b) Argon, potassium, and calcium are called isobars, not isotopes. Construct an argument — using specific values of Z and A — distinguishing why these three atoms cannot be isotopes of the same element, and what structural feature they share that makes them isobars.

Answer section

Look for

- Correctly identifies that both carbon isotopes have $Z=6$ and therefore the same electronic configuration (2,4), which determines chemical behaviour — chemical properties depend on electron arrangement, not nuclear mass.
- Correctly distinguishes: isotopes require the same Z ; argon ($Z=18$), potassium ($Z=19$), and calcium ($Z=20$) all have different atomic numbers, so they cannot be isotopes.
- Correctly identifies the isobar feature: all three have $A=40$ (same total number of nucleons) but different atomic numbers — same mass number, different identities.
- Argument uses specific numerical values of Z and A , not just the definitions.
- Reasoning is logically sequenced: the student moves from the definition to the specific values to the conclusion, not from the conclusion backward.